

Process Characterization Report

A-Mab Drug Substance — Production Bioreactor (Step 3) — PCR-003

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

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Field	Value
Document ID	PCR-003
Document class	Process Characterization Report
Title	Process Characterization Report: Production Bioreactor (Step 3)
Product	A-Mab
Version	1.0
Effective date	2026-07-24
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor / sites	Novacyte Biologics; Novacyte Biologics — Cambridge, MA (Development) → Novacyte Biologics — Grafton, WI (Commercial DS)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

2-AB, 2-aminobenzamide; AMV, analytical method validation; ANOVA, analysis of variance; Cpk, process capability index (one sided); CPP, critical process parameter; CQA, critical quality attribute; CV, coefficient of variation; GPP, general process parameter; HCP, host cell protein; HILIC, hydrophilic interaction liquid chromatography; HMW, high molecular weight; icIEF, imaged capillary isoelectric focusing; IgG1, immunoglobulin G subclass 1; KPP, key process parameter; LOQ, limit of quantitation; MSAT, Manufacturing Science and Technology; NOR, normal operating range; PAR, proven acceptable range; pCO₂, dissolved carbon dioxide partial pressure; qPCR, quantitative polymerase chain reaction; RMSE, root mean square error; RSD, relative standard deviation; RSM, response surface methodology; SEC, size exclusion chromatography; SOP, standard operating procedure; UPLC, ultra performance liquid chromatography; VCD, viable cell density; WC-CPP, well-controlled critical process parameter.

Executive summary

The production bioreactor is characterized, and the multivariate region defined in this report supports the operating ranges proposed for commercial manufacture. A-Mab is produced by fed-batch culture of a humanized IgG1 cell line at a working volume of 15,000 L. The step establishes the glycan, charge variant and aggregate attributes of the drug substance, so it carries most of the product-quality risk in the upstream process. The studies were executed on a qualified scale-down model under the paired protocol PCP-003, following the lifecycle approach to process validation (U.S. Food and Drug Administration 2011), the enhanced development approach of ICH Q8 and ICH Q11 (International Council for Harmonisation 2009, 2012), and the quality-attribute and risk framework of the A-Mab case study (CMC Biotech Working Group 2009).

The study carried 9 parameters, on the risk basis established in RA-001. Of those, 5 were studied multivariately, in a screening design of 19 runs followed by a response-surface design of 28 runs, and the remaining 4 were assessed one at a time. The screening design identified the active factors. The response-surface model is the predictive model and the basis of the design space, and it accounts for between 91.1% and 97.1% of the observed variation in the 5 measured attributes, with lack of fit not significant for any of them.

One corner of the characterized region is excluded from the design space. With culture pH, culture duration and pCO₂ all at their low edges and culture temperature at its high edge, the fitted model predicts galactosylation of 40.27 %, which is 0.27 percentage points above the upper acceptance limit of 40 %. That exclusion covers 3 of the 194,481 grid points examined across the four response-surface factors. Across the 16 corners of the normal operating ranges the highest predicted value of the same attribute is 31.8 %, which leaves 8.2 percentage points to the limit. The design space is the characterized region with that one corner removed, and the normal operating ranges sit well inside it.

Commercial-scale capability was estimated from 2,000 simulated batches with every parameter varying inside its normal operating range. All seven attributes the step sets meet their acceptance criteria. High Mannose carries the tightest capability, with a one-sided Cpk of 1.94 and 2.99 percentage points between the predicted mean and the binding lower limit. Two deviations were recorded during execution (§13). Both were investigated, impact assessed and retained, and neither changed a parameter classification or the boundaries of the operating region.

The step does not control host cell protein, residual DNA or aggregate at the drug substance on its own. Those attributes are generated in the culture and are then reduced by the purification train, and the clearance credited to the downstream steps is reported in PCR-005, PCR-007 and PCR-008. The capability figures given here rest on a qualified scale-down model with all parameters held inside their normal

operating ranges. They are not a commercial-scale confirmation, which is the object of the process performance qualification. This report rolls up into PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody produced in a recombinant mammalian cell line. Its primary mechanism of action is antibody dependent cellular cytotoxicity, so the N-linked glycan profile of the molecule is directly linked to efficacy and is treated as a set of critical quality attributes throughout development (CMC Biotech Working Group 2009). The drug substance process is a platform process, and its unit operations and their principal roles are given in Table 1.1.

Table 1.1: The A-Mab drug substance process train, with the principal role of each unit operation in the control strategy.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

The production bioreactor is the only step of the drug substance process at which product quality attributes are formed. The glycosylation and charge variant distributions are established inside the cell and in the culture fluid, and the platform purification train does not modify them. Protein A capture, cation exchange and anion exchange chromatography separate on affinity and on charge, and none of them discriminates between the glycoforms of an intact antibody. What the downstream steps do is remove process impurities and reduce the aggregate that leaves the bioreactor, and that clearance is characterized in the reports for those steps.

The step is operated as a fed-batch culture at a working volume of 15,000 L, with pH, temperature and dissolved oxygen under closed-loop control, pCO₂ monitored continuously, and nutrient feed delivered on a fixed schedule. At the nominal set-point the culture reaches a peak viable cell density of 13.5×10^6 cells/mL and is harvested at a viability of 67 % after 17 days, giving a titre of 4.37 g/L and 66 kg of antibody in the harvested culture fluid. The five attributes measured across the characterization studies are afucosylation, galactosylation, high mannose, acidic charge variants and high molecular weight aggregate. Host cell protein and residual DNA are also generated here, and both are governed at the drug substance by downstream clearance as much as by culture conditions.

1.2 Objectives

This report presents the executed characterization of the production bioreactor against the objectives set in PCP-003. Those objectives were:

- (i) to quantify the relationship between each studied process parameter and the quality attributes the step forms, across ranges wider than the intended operating ranges;
- (ii) to define a multivariate operating region over the well-controlled culture parameters in which every attribute in scope remains within its acceptance criterion;
- (iii) to establish proven acceptable ranges for each parameter against each attribute it governs, both at the set-point and under normal operating variation of the other parameters;
- (iv) to classify every parameter carried into the study according to the decision logic of SOP-4001, on the basis of the demonstrated effects and the risk of leaving the operating region; and
- (v) to justify the ranges and controls carried forward into Stage 2 and into the overall control strategy consolidated in PCMR-001.

1.3 Regulatory and scientific basis

The study is a Stage 1 process design activity in the sense of the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). It follows the enhanced development approach described in ICH Q8 and ICH Q11, in which prior knowledge and a formal risk assessment direct the experimental work, and the outcome is an operating region and a control strategy rather than a set of fixed points (International Council for Harmonisation 2009, 2012). Criticality is treated as a continuum, as described in ICH Q9 and applied throughout the A-Mab case study, so an attribute is assigned a level from very low to very high and the attributes at high and very high criticality are called critical (International Council for Harmonisation 2023b; CMC Biotech Working Group 2009).

No viral clearance claim is made for this step. The viral safety evaluation is modular and rests on the low-pH inactivation, anion exchange and virus filtration steps, and it

is assessed under ICH Q5A in PCR-006, PCR-008 and PCR-009 (International Council for Harmonisation 2023a). Nothing in this report contributes to that claim.

1.4 Related documents

Table 1.2 lists the documents this report depends on and the document it rolls up into. PCP-003 is the approved protocol for the work reported here, and the two documents are read together.

Table 1.2: Related documents in the A-Mab characterization package.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-003	Process Characterization Plan (Production Bioreactor (Step 3))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The fed-batch platform used for A-Mab has an extensive performance history. It has supported three previous humanized IgG1 antibodies (X-Mab, Y-Mab and Z-Mab) through clinical supply and commercial manufacture, and the medium, feed strategy and control scheme carried into A-Mab are the platform ones (CMC Biotech Working Group 2009). The consequence for this study is that the characterization confirms and bounds mechanisms that are already understood, instead of discovering them. The acceptance limits for aggregate and for host cell protein are themselves set from prior clinical experience with X-Mab.

The platform establishes what each culture parameter is expected to do. Culture pH sets the intracellular and Golgi pH, which governs the activity of the glycosyltransferases and the fucosyltransferase, so pH is expected to move afucosylation, galactosylation and high mannose together. Culture temperature changes cell-specific productivity and the residence time of the molecule in the secretory pathway, so a lower temperature is expected to raise terminal galactosylation. Dissolved carbon dioxide acidifies the intracellular compartment and is expected to suppress galactosylation and, through the same route, the deamidation that produces acidic charge variants. Osmolality acts on cell-specific productivity and on the same enzymes, but more weakly. Culture duration acts through the decline in viability that occurs late in a fed-batch culture, which releases intracellular enzymes and host cell protein into the culture fluid and raises aggregate.

Two of these expectations are not independent. pH and duration act on the same glycosylation machinery at different points in the culture, so an interaction between them was anticipated and was the main reason for taking the study beyond one factor at a time. The study was designed to confirm or refute each of these expectations (§5.2), and §5.4 returns to them against the fitted models.

2.2 Quality attributes in scope

The attributes this step sets are listed in Table 2.1 with their acceptance criteria and their assigned criticality. The Tool #1 score is the product of impact and uncertainty, and the attributes scored high or very high are treated as critical (CMC Biotech Working Group 2009).

Table 2.1: Quality attributes set by the production bioreactor, with acceptance criteria and assigned criticality.

CQA	Category	Acceptance	Criticality	Tool #1
Afucosylation	glycosylation	2-13 % of glycans	H	60
Galactosylation (total %Gal)	glycosylation	10-40 % (G1+G2 equiv.)	H	48
High Mannose	glycosylation	3-10 % of glycans	VH	80
Aggregates (HMW)	size variant	0-5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18-40 % (CEX/icIEF)	VL	4
Host Cell Protein (HCP)	process impurity	0-100 ng/mg	M-H	36
Residual DNA	process impurity	0-0.001 ng/dose	VL	6

High mannose carries the highest score in the table, because its impact on clearance and its residual uncertainty are both high, and it is the only attribute of the seven at very high criticality. Afucosylation and galactosylation follow, both linked to potency through antibody dependent cellular cytotoxicity and complement dependent cytotoxicity. Aggregate is critical on immunogenicity grounds. Acidic charge variants and residual DNA score low, which reflects a well characterized structure-function relationship in the first case and a very large clearance margin in the second. Host cell protein sits between the two groups.

Three of the seven attributes are governed by more than one step. Aggregate leaving the bioreactor is raised slightly during the low-pH hold and is then polished by cation exchange, so the drug substance level reported in §8 is not attributable to culture conditions alone (PCR-006, PCR-007). Host cell protein and residual DNA are generated in the culture at levels several orders of magnitude above the drug substance criterion and are cleared by Protein A, cation exchange and anion exchange (PCR-005, PCR-007, PCR-008). This report characterizes the contribution the bioreactor makes to each of them, which is the input level, and it makes no clearance claim.

2.3 Risk-based prioritization and parameter selection

RA-001 assessed every parameter of this step before any characterization run was executed, using the risk ranking and filtering method of the A-Mab case study. Each parameter was scored on its potential impact on a quality attribute and on its potential to interact with other parameters, and the score determined the type of study assigned to it (CMC Biotech Working Group 2009; International Council for Harmonisation

2023b). Parameters with a credible route to a quality attribute and a credible interaction went into the multivariate design. Parameters whose expected effect is on process performance and not on product quality were assigned to univariate study. No parameter of this step was left unstudied.

The assignment is reproduced in the classification table in §9, and the outcome was 5 multivariate parameters (culture pH, culture temperature, pCO₂, osmolality and culture duration) and 4 univariate ones (dissolved oxygen, initial viable cell concentration, basal medium concentration and nutrient feed-1 volume). The five multivariate parameters are the ones the risk assessment linked to the glycan, charge variant and aggregate attributes, and they are also the parameters for which prior knowledge predicts interactions.

The ranges studied are wider than the ranges the process will be run in. Each multivariate characterization range is between 1.6 and 2.5 times the width of the corresponding normal operating range, which follows the practice of carrying the study well beyond the control range so that the edges of the region are probed and not assumed (CMC Biotech Working Group 2009). Two constraints bounded the expansion. The low pH edge was set where the platform cell line begins to lose viability late in culture, and the high duration edge was set by the harvest criteria in SOP-2003, since a culture cannot be held past the point at which viability falls below the harvest limit. The resulting ranges are the knowledge space for this step, and they are listed in Table 4.1.

3 Materials and methods

3.1 Scale-down model and its qualification

All characterization runs were executed on a qualified scale-down model of the commercial bioreactor, and the warrant for applying the results to commercial scale rests on that qualification. The scale-down system is a stirred-tank bioreactor of the same geometry and with the same control capability as the commercial vessel, covering pH, dissolved oxygen, temperature, agitation and nutrient addition. The scale-independent parameters (pH, temperature, dissolved oxygen, initial viable cell concentration and culture duration) were operated at the values proposed for commercial operation. The scale-dependent parameters (agitation rate, gas flow rates, headspace pressure and working volume) were set so that the scale-down system matched commercial-scale performance and not commercial-scale settings, which is the standard approach for a stirred-tank culture and is the approach described in the case study (CMC Biotech Working Group 2009).

Qualification followed SOP-1001. Replicate set-point runs on the scale-down system were compared against commercial-scale engineering batches on the process performance measures (peak viable cell density, harvest viability, titre) and on the full quality panel, and the comparison and its acceptance criteria are reported in PCP-003. The model is qualified for the ranges characterized here and for the cell line and medium used here. It is not qualified for a change of medium platform, and any such change would require the comparison to be repeated.

The centre-point replicates executed within each design serve a second purpose. They give an estimate of pure error that is independent of the fitted model, which is what makes the lack-of-fit test in §5.3 a real test and not a restatement of the residual variance. Their reproducibility is reported in §5.1.

3.2 Cell line, media and culture operation

The production culture was operated per SOP-2003. Cells were expanded through the seed train, inoculated into the production vessel at the target initial viable cell concentration, and cultured in fed-batch mode with nutrient feed-1 delivered on the platform schedule. Basal medium and feed were prepared per SOP-2004 from released lots. Culture pH was controlled by carbon dioxide sparge and base addition, temperature by the vessel jacket, and dissolved oxygen by a gas cascade (SOP-2003). Harvest was triggered by the culture duration under study, subject to the viability floor in SOP-2003.

Inputs that are controlled to a platform specification were not characterized here. The working cell bank, the basal medium formulation, the feed formulation and the gas supplies are identity tested and released against their specifications, and their control is a control on materials and not a process parameter. The nominal fed-batch profile at the set-point is shown in Figure 3.1, and it is the reference against which each designed run was assessed for normal culture behaviour.

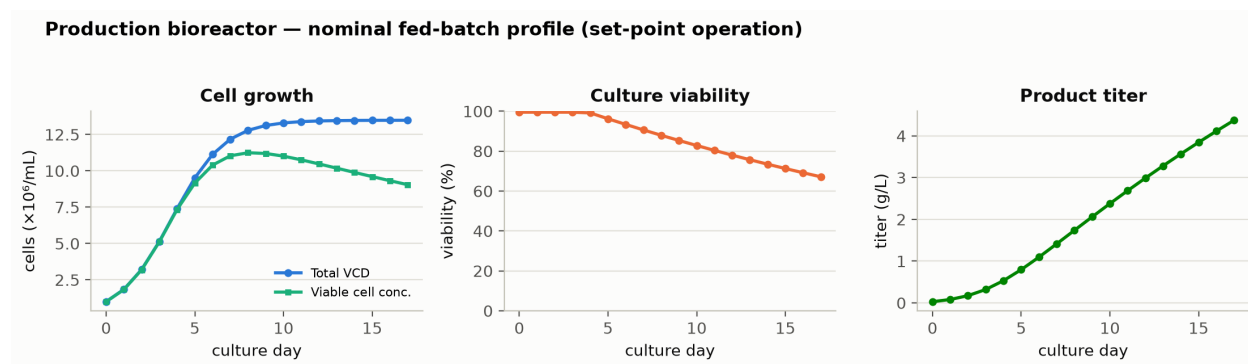


Figure 3.1: Nominal production-bioreactor profile at the set-point: viable cell density, culture viability and product titre over the fed-batch culture.

3.3 Analytical methods

Every attribute reported here was measured by a validated method. Table 3.1 lists the procedures and method validation reports that apply to this step.

Table 3.1: Controlled procedures and validated analytical methods applied to the production bioreactor characterization.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2003	Operation of the Production Bioreactor (Fed-Batch)	SOP
SOP-2004	Cell-Culture Media and Feed Preparation	SOP
SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC	SOP
SOP-3011	Size-Variant Analysis by SEC-HPLC	SOP
SOP-3012	Host-Cell-Protein Quantitation by ELISA	SOP
SOP-3013	Charge-Variant Analysis by icIEF	SOP
SOP-3014	Residual Host-Cell DNA by qPCR	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation

Reference	Title	Type
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation

The three glycan attributes were measured on a single released N-glycan map by 2-AB labelled HILIC-UPLC (SOP-3010, AMV-3010), which reports afucosylation, galactosylation and high mannose from one injection. Size variants were measured by SEC-HPLC (SOP-3011, AMV-3011) and are reported as percent high molecular weight. Charge variants were measured by icIEF (SOP-3013, AMV-3013) and are reported as the summed acidic peak area. Host cell protein was measured by a platform ELISA (SOP-3012, AMV-3012) and residual DNA by qPCR (SOP-3014, AMV-3014). Method performance is summarized in Appendix D, and the full validation data are in the AMV reports.

Assay precision matters to the interpretation of the smaller effects. The glycan map has a precision of 2.5 % RSD and the host cell protein ELISA a precision of 9.5 % RSD, so the assay resolves a fraction of a percentage point on any of the three glycan attributes at the levels they are reported at. Reproducibility between runs is the wider quantity, since it carries the biological variation of the culture as well as the analytical variation. A shift of about one percentage point in a glycan attribute therefore becomes interpretable through replication or through a fitted model, and not from a single determination. §5.1 separates the analytical contribution from the process contribution using the centre-point replicates.

3.4 Sampling plan

Each designed run was sampled daily for viable cell density, viability, metabolites and titre, which supports the assessment of normal culture behaviour but is not part of the designed analysis. The characterization responses were measured on the harvested culture fluid of each run, taken at the culture duration assigned to that run and clarified before analysis. One harvest sample per run was tested against the full panel, so each run contributes one observation per response.

Centre-point runs were replicated within each design. The screening design carried 3 centre points and the response-surface design carried 4, all executed at the set-point of every factor. Replicating at the centre instead of at a corner keeps the pure-error estimate at the condition the process will actually run at, and it is the condition for which prior process history gives an independent expectation.

3.5 Statistical methods

All factors were analysed on a coded scale, in which the set-point of a factor is zero and the edges of its characterization range are minus one and plus one. Coding puts the estimates on a common footing, so that an effect is the change in the response across the full characterization range of that factor and effects can be compared between parameters with different units (CMC Biotech Working Group 2009).

The two designs answer different questions and are analysed differently. The screening design was analysed by ordinary least squares with all main effects and all two-factor interactions, and an effect is reported as twice the fitted coefficient on the coded factor. Its purpose is to identify which factors are active. The response-surface design was analysed with a full quadratic model in the factors carried forward, which adds the squared terms needed to describe curvature, and it is the model used for prediction, for the design space and for the proven acceptable ranges. A significance level of $\alpha = 0.05$ was used throughout.

Model adequacy was assessed on four measures (SOP-1002). R^2 and adjusted R^2 describe how much of the observed variation the model accounts for, with the adjustment penalising terms that do not earn their degrees of freedom. Predicted R^2 , computed from the PRESS statistic by leaving each run out in turn, describes how the model performs on data it has not seen. The overall F test asks whether the model as a whole explains more than noise. Finally, the residual variance was partitioned into lack of fit and pure error using the centre-point replicates, and a significant lack of fit would mean the model form is wrong even where its coefficients are well estimated. A factor with no significant term for a given response is treated as having no demonstrated effect on that response over the range studied.

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches, with every parameter drawn independently within its normal operating range and the fitted models used to predict each attribute. The capability index is one sided and is reported against the binding acceptance limit for each attribute.

Proven acceptable ranges were computed for each attribute against each response-surface factor, under two analyses. The first holds the other factors at their set-points and evaluates the fitted model across the parameter's characterization range. The second varies the other factors within their normal operating ranges by Monte-Carlo of the same model, and requires the 95 % predictive interval of the attribute to stay within acceptance, so it is the more demanding of the two. Both are reported in §7.

4 Study design

4.1 Factors, ranges and the knowledge space

The design covers 9 parameters, and Table 4.1 gives the set-point, normal operating range, characterization range, study type and final classification of each. The classification column is an outcome of this study and is argued parameter by parameter in §9. The characterization range is the range over which the parameter was actually varied, and it is the edge of the knowledge space, not a proven acceptable range. The proven acceptable ranges are computed quantities and are given in §7.

Table 4.1: Production-bioreactor parameters, ranges, study type and final classification.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Culture pH	pH	6.85	6.75–6.95	6.6–7.1	WC-CPP	multivariate
Culture temperature	°C	35	34.5–35.5	34–36	WC-CPP	multivariate
Dissolved CO ₂ (pCO ₂)	mmHg	100	70–130	40–160	WC-CPP	multivariate
Osmolality	mOsm	400	375–425	360–440	WC-CPP	multivariate
Culture duration	days	17	16–18	15–19	WC-CPP	multivariate
Dissolved oxygen	% sat	50	40–60	30–70	KPP	univariate
Initial viable cell conc.	e6 cells/mL	1	0.8–1.2	0.5–1.5	KPP	univariate
Basal medium concentration	X	1.2	1–1.4	0.8–1.6	GPP	univariate
Nutrient feed-1 volume	% WV	12	10–14	9–15	KPP	univariate

The five multivariate factors are coded A to E for the analysis, and Table 4.2 gives the mapping together with the design each factor appears in.

Table 4.2: Coded factor legend for the production-bioreactor designs.

Code	Factor	Unit	Studied in
A	Culture pH	pH	screening + RSM
B	Culture temperature	°C	screening + RSM
C	Dissolved CO ₂ (pCO ₂)	mmHg	screening + RSM
D	Osmolality	mOsm	screening
E	Culture duration	days	screening + RSM

Each range was set from prior knowledge and not by a fixed multiple. The pH range of 6.6 to 7.1 spans the region in which the platform cell line grows and produces normally, and it is the widest range the harvest viability criterion allows. The temperature range of 34 to 36 °C is set by the control capability of the commercial vessel, which holds the culture well inside that band and makes a wider excursion implausible. The pCO₂ range of 40 to 160 mmHg covers the accumulation observed at commercial scale, where sparge efficiency falls with vessel size and the upper part of the range is the realistic one. The osmolality range of 360 to 440 mOsm covers the drift produced by feeding and base addition over a full culture. The duration range of 15 to 19 days brackets the harvest window either side of the set-point.

4.2 Screening design

The screening study was a two-level design in the 5 multivariate factors, executed as a half fraction of resolution V with 3 centre points, for 19 runs in total. The generator sets culture duration as the product of the other four factors, so the defining relation involves all five. At that resolution the main effects are aliased only with four-factor interactions and the two-factor interactions only with three-factor interactions, and effects of that order are not credible for a culture process. The design therefore estimates all 5 main effects and all ten two-factor interactions clear of one another. The runs were randomised in execution order (SOP-1002). The full design matrix with responses is given in Appendix A.

The 16 factorial runs support a model of 16 terms, so the residual degrees of freedom come from the centre points alone. This is a deliberate property of a screening design and it is the reason the screening fit is not used for prediction. §5.2 reports what it does establish.

4.3 Response-surface design

The response-surface study was a face-centred central composite design in the 4 factors carried forward from screening (culture pH, culture temperature, culture duration and pCO₂), with 4 centre points, for 28 runs in total. The face-centred form places the axial runs on the faces of the cube and not outside it, so no run is

executed beyond the characterization range of any factor. That choice keeps every prediction inside the region actually run, at the cost of a less precise estimate of the pure quadratic terms.

Osmolality was not carried into the response-surface design. Its screening main effect is the smallest of the five on galactosylation and on high mannose, which are the two responses that drive the design space and the capability claim, and osmolality is held close to its set-point in routine operation by the feed and base strategy. It was operated at its set-point in every response-surface run, so the response-surface model, the design space in §6 and the proven acceptable ranges in §7 all apply at osmolality within its normal operating range. The parameter does appear in three significant galactosylation interactions in the screening study, the largest of which is the interaction with culture temperature at +3.70 percentage points. That term is bounded by the screening estimate and by holding osmolality within its normal operating range, and §9 classifies the parameter on that basis. The full response-surface matrix with responses is given in Appendix B.

4.4 Univariate assessment

Four parameters were assessed one at a time: dissolved oxygen, initial viable cell concentration, basal medium concentration and nutrient feed-1 volume. RA-001 ranked each of them against process performance and not against a quality attribute, which is what assigned them to single-factor study instead of the designed experiment. Each was operated across its characterization range with all other parameters at their set-points, and the harvest material was released against the same analytical panel.

The univariate work supports the operating ranges of those four parameters for process performance. The results of the individual runs are held in the study record referenced by PCP-003 and are not reproduced here, since no effect on a quality attribute was demonstrated for any of the four over the range studied. The univariate work does not extend the quality design space, which rests entirely on the multivariate studies, and it is not used in §6 or §7. The classifications that follow from it are given in §9.

5 Results

5.1 Centre-point performance and reproducibility

Reproducibility at the centre point is good for three of the five responses and moderate for galactosylation and high mannose. Table 5.1 gives the mean, standard deviation and coefficient of variation of the 4 response-surface centre points, which are the pure-error basis for the lack-of-fit tests in §5.3.

Table 5.1: Centre-point reproducibility of the response-surface design (n = 4 replicates at the set-point of every factor).

Response	n	Mean	SD	%CV
Afucosylation	4	7.61	0.355	4.66
Galactosylation	4	25.9	2.34	9.02
High mannose	4	5.66	0.559	9.88
Acidic variants	4	21.5	0.877	4.08
Aggregates (HMW)	4	1.51	0.0985	6.53

High mannose is the least reproducible response, at 9.9 % CV, and acidic variants the most reproducible, at 4.1 %. Galactosylation follows close behind at 9.0 %, and that spread is the pure error the galactosylation lack-of-fit test uses. Galactosylation is the response that defines the design space, so its variation at the centre point is the term that test finally rests on. The glycan map contributes 2.5 % RSD of that variation (§3.3), so most of the spread in high mannose is biological in origin and not analytical. This matters for the interpretation of the smaller effects, and it is the reason a shift of about one percentage point in a glycan attribute is reported here as an effect only when the fitted model resolves it.

In the screening centre points, executed earlier in the campaign, the least reproducible response is again high mannose, at 11.5 % CV. The ordering of the other four differs. Galactosylation is the most reproducible response in that set, at 1.8 %, against 9.0 % at the response-surface centre points. Each figure rests on a handful of replicates, so the difference belongs to the two replicate sets and not to the assay, and the lack-of-fit tests that support the design space use the response-surface set.

5.2 Screening: factor effects

The screening design separates the active factors from the inactive ones, and it does so with different force from one response to another. Table 5.2 summarizes the fitted screening models.

Table 5.2: Screening model summary. The high R^2 values reflect a near-saturated design; predicted R^2 is negative for every response, which is why these models are not used for prediction.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Afucosylation	19	0.965	0.79	-47.4	5.51	0.0926	1.06
Galactosylation	19	0.998	0.988	-1.33	103	0.00138	0.773
High mannose	19	0.942	0.65	-7.99	3.23	0.182	0.619
Acidic variants	19	0.994	0.963	-0.442	32.2	0.00766	0.957
Aggregates (HMW)	19	0.995	0.972	-2.44	42.2	0.00515	0.0829

The screening models fit the executed runs closely, and they have almost no predictive latitude. Each model estimates 16 terms from 19 runs, which leaves 3 residual degrees of freedom, and predicted R^2 is negative for all 5 responses (the highest value is -0.44). A negative predicted R^2 means the model predicts a held-out run worse than the response mean would. The overall F test divides the panel as well. It is significant for galactosylation, acidic variants and aggregates, and it does not reach significance for afucosylation and high mannose, so the individual terms of those models are read as indicative and are confirmed against the response-surface design in §5.3. The screening design identifies which factors matter. The response-surface model is the predictive model and the basis of the design space, and the effect magnitudes reported below are refined by it in §5.3.

Table 5.3 lists every screening term significant at $\alpha = 0.05$, by response.

Table 5.3: Screening terms significant at $\alpha = 0.05$, by response. An effect is the change in the response across the full characterization range of the factor. Factor codes are given in Table 4.2.

Response	Term	Effect	p-value
Afucosylation	AE	2.46	0.0191
Afucosylation	E	-2.37	0.0209
Afucosylation	AC	-1.76	0.0455
Galactosylation	E	-7.82	0.000264
Galactosylation	AE	6.9	0.000382
Galactosylation	C	-5.79	0.000644
Galactosylation	A	-4.56	0.00131
Galactosylation	BD	3.7	0.00242
Galactosylation	AC	3.6	0.00261

Response	Term	Effect	p-value
Galactosylation	B	3.54	0.00274
Galactosylation	AB	3.28	0.00343
Galactosylation	D	-2.45	0.0079
Galactosylation	CD	2.18	0.011
Galactosylation	DE	1.93	0.0155
Galactosylation	BE	1.49	0.0306
High mannose	A	1.51	0.0164
High mannose	B	-1.39	0.0205
Acidic variants	C	-8.15	0.000441
Acidic variants	A	-4.17	0.00319
Acidic variants	B	2.97	0.00848
Acidic variants	D	-2.55	0.0129
Acidic variants	AB	1.98	0.0255
Acidic variants	AC	-1.93	0.0275
Acidic variants	BE	1.73	0.0364
Aggregates (HMW)	A	0.834	0.000268
Aggregates (HMW)	E	0.578	0.0008
Aggregates (HMW)	BE	-0.141	0.0424

Galactosylation is the most responsive attribute in the panel and the most heavily interacting. Culture duration carries the largest single effect, -7.8 percentage points across its characterization range, and the pH by duration interaction is almost as large at +6.9. pCO₂ and culture pH follow, at -5.8 and -4.6. Of the 15 estimable terms, 12 are significant, which is the clearest evidence in the study that a one-factor-at-a-time treatment of this attribute would be misleading.

Acidic charge variants are governed mainly by pCO₂, whose effect of -8.2 percentage points is the largest single effect anywhere in the screening study. Culture pH and culture temperature follow at -4.2 and +3.0, and osmolality contributes -2.6. The pCO₂ direction is the one §2.1 predicted, and the effect is negative as expected. §2.1 sets no direction for pH, temperature or osmolality on this attribute, and the fitted pH direction is not the one deamidation chemistry gives on its own (§5.4). The attribute is of very low criticality and its acceptance range is wide, so these are large effects on a response that has room to absorb them.

Afucosylation behaves differently, and the difference is informative. Only three of its terms are significant, and the largest of them is the pH by duration interaction at +2.5 percentage points, followed by culture duration at -2.4. The main effect of pH is not significant at $\alpha = 0.05$ ($p = 0.060$), yet pH appears in two of the three significant terms. The effect of pH on afucosylation depends on how long the culture is held, so the main effect on its own understates it. Aggregate is the most straightforward of the five mechanistically, with culture pH at +0.83 percentage points and culture duration at +0.58 against an acceptance limit of 5 %. The duration effect is positive, which is the direction §2.1 expected from the decline in viability late in culture.

High mannose is the sparsest response, and the null results are as useful as the significant ones. Only culture pH and culture temperature have a significant effect,

at +1.5 and -1.4 percentage points. pCO₂, osmolality and culture duration show no significant effect on high mannose over the ranges screened, and no interaction involving them is significant either. The screening model for this response does not reach significance overall, so those null results are carried into the response-surface design for confirmation before they are relied on. pCO₂ and culture duration are retained in the knowledge space and are classified in §9 on the evidence from both designs. Osmolality was not carried into the response-surface design, so its null result on this attribute rests on the screening study alone (§4.3). High mannose is the only attribute of very high criticality at this step, which is why §5.3 examines its response surface in full.

Dissolved oxygen was held at its set-point throughout the designed runs and was not varied. Platform manufacturing experience has established that dissolved oxygen does not influence the glycan profile at any scale, so the factor was not carried into the multivariate work and no further study of it is required.

Taken together, the 19 screening runs establish the multivariate design space for the well-controlled culture parameters, and they confirm that the estimated effects hold across the operating combinations the design did not itself execute.

5.3 Response-surface models

A full quadratic model was fitted to each of the 5 responses over the four factors carried into the response-surface design. Table 5.4 summarizes model adequacy.

Table 5.4: Response-surface model summary. Each model carries 15 terms fitted to 28 runs.

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Afucosylation	28	0.963	0.923	0.795	24.3	5.05e-07	0.584
Galactosylation	28	0.971	0.94	0.915	31.3	1.06e-07	1.36
High mannose	28	0.911	0.815	0.65	9.48	0.000118	0.472
Acidic variants	28	0.938	0.872	0.598	14.1	1.27e-05	1.57
Aggregates (HMW)	28	0.969	0.936	0.835	29.3	1.58e-07	0.0985

The models are adequate for all five responses and each of them retains predictive value on runs it has not seen. R² ranges from 0.91 to 0.97, adjusted R² does not fall below 0.81 (high mannose), and predicted R² does not fall below 0.60 (acidic variants), against a highest value of 0.92. The response-surface model is the predictive model and the basis of the design space, and it supersedes the screening estimates wherever the two differ. Its predictions are mean levels, so a prediction at the edge of the region carries about a one in two chance of falling on either side of the predicted value if the variability around the mean is symmetric (CMC Biotech Working Group 2009). The design space in §6 is therefore not offered as an assurance statement on its own, and

the assurance that a batch meets its criteria comes from the capability analysis in §8.

Lack of fit is not significant for any response. The residual variance was partitioned against the 3 pure-error degrees of freedom from the centre points, and the smallest lack-of-fit p-value across the five responses is 0.15 (acidic variants), well above $\alpha = 0.05$. The quadratic form is therefore sufficient to describe the surfaces over the ranges run, and no evidence was found that a higher-order model is needed. Table 5.5 and Table 5.6 give the partition for the two responses that drive the design space and the capability claim.

Table 5.5: Analysis of variance with lack-of-fit partition, galactosylation response-surface model.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	811	14	57.9	31.3	1.06e-07
Residual	24.1	13	1.85	nan	nan
Lack of fit	7.69	10	0.769	0.141	0.992
Pure error	16.4	3	5.46	nan	nan

Table 5.6: Analysis of variance with lack-of-fit partition, high mannose response-surface model.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	29.6	14	2.11	9.48	0.000118
Residual	2.9	13	0.223	nan	nan
Lack of fit	1.96	10	0.196	0.628	0.748
Pure error	0.938	3	0.313	nan	nan

The coefficient tables for the same two responses are given in Table 5.7 and Table 5.8, and the remaining three are in Appendix C. For galactosylation, the four main effects, the pH by temperature, pH by duration and pH by pCO₂ interactions are all significant, and none of the quadratic terms is. The surface is close to a plane with strong twist, and the twist is what excludes the corner identified in §6. For high mannose the picture is much simpler, with significant terms for culture pH, culture temperature and two pH interactions, and no significant curvature.

Table 5.7: Response-surface coefficients on coded factors, galactosylation.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	25.8	0.47	54.8	9.11e-17	***
A	-2.21	0.321	-6.89	1.09e-05	***
B	2.3	0.321	7.18	7.13e-06	***
E	-3.82	0.321	-11.9	2.27e-08	***
C	-2.43	0.321	-7.56	4.11e-06	***

Term	Coef.	Std. err.	t	p-value	Sig.
AB	1.67	0.34	4.92	0.000279	***
AE	2.72	0.34	7.99	2.28e-06	***
AC	2.35	0.34	6.9	1.09e-05	***
BE	-0.217	0.34	-0.639	0.534	
BC	0.315	0.34	0.928	0.37	
EC	0.332	0.34	0.975	0.347	
A ²	-0.837	0.847	-0.988	0.341	
B ²	0.378	0.847	0.446	0.663	
E ²	0.937	0.847	1.11	0.289	
C ²	-0.373	0.847	-0.441	0.667	

Table 5.8: Response-surface coefficients on coded factors, high mannose.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	5.89	0.163	36	2.06e-14	***
A	1.09	0.111	9.77	2.36e-07	***
B	-0.421	0.111	-3.78	0.00227	**
E	-0.188	0.111	-1.69	0.115	
C	0.212	0.111	1.91	0.0788	
AB	0.148	0.118	1.26	0.231	
AE	-0.29	0.118	-2.46	0.0289	*
AC	0.293	0.118	2.48	0.0276	*
BE	0.0874	0.118	0.74	0.472	
BC	-0.118	0.118	-0.996	0.337	
EC	-0.0402	0.118	-0.341	0.739	
A ²	0.116	0.294	0.393	0.701	
B ²	0.094	0.294	0.32	0.754	
E ²	-0.215	0.294	-0.731	0.478	
C ²	0.126	0.294	0.43	0.674	

Figure 5.1 shows the fitted surfaces over the two factors that carry the largest interaction, with the other two held at their set-points. The pH by duration twist is visible in the afucosylation and galactosylation panels, where the direction of the pH effect changes with culture duration. High mannose rises monotonically with pH and is almost flat in duration, which is the graphical form of the null result reported in §5.2. Aggregate rises with both factors and stays far below its limit throughout.

Response-surface predictions (remaining factors held at set-point)

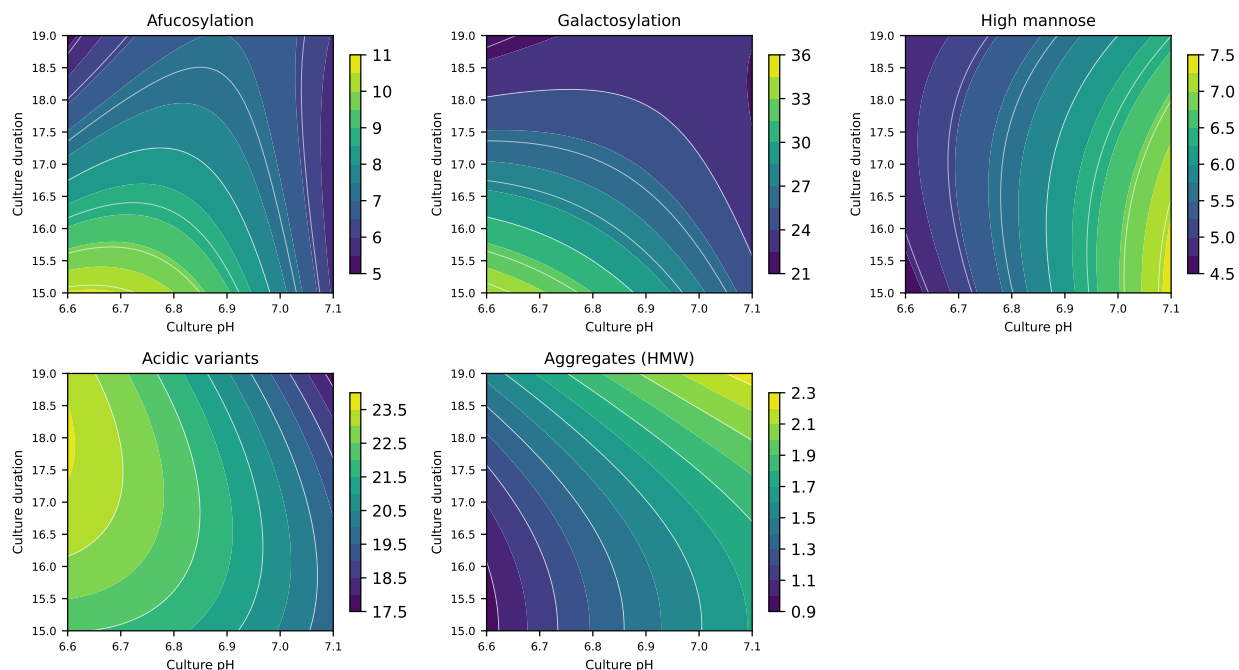


Figure 5.1: Fitted response surfaces for the five measured attributes over culture pH and culture duration, with culture temperature and pCO₂ held at their set-points.

Model diagnostics for the two responses that carry the argument are given in Figure 5.2 and Figure 5.3. In both cases the standardized residuals scatter without structure about zero across the fitted range, the normal quantile plot is close to a straight line, and the actual against predicted plot lies on the identity line. Nothing in the diagnostics suggests a missing term, a variance that changes with the level of the response, or an influential run. Diagnostics for the remaining three responses were reviewed to the same standard and are held in the study record referenced by PCP-003.

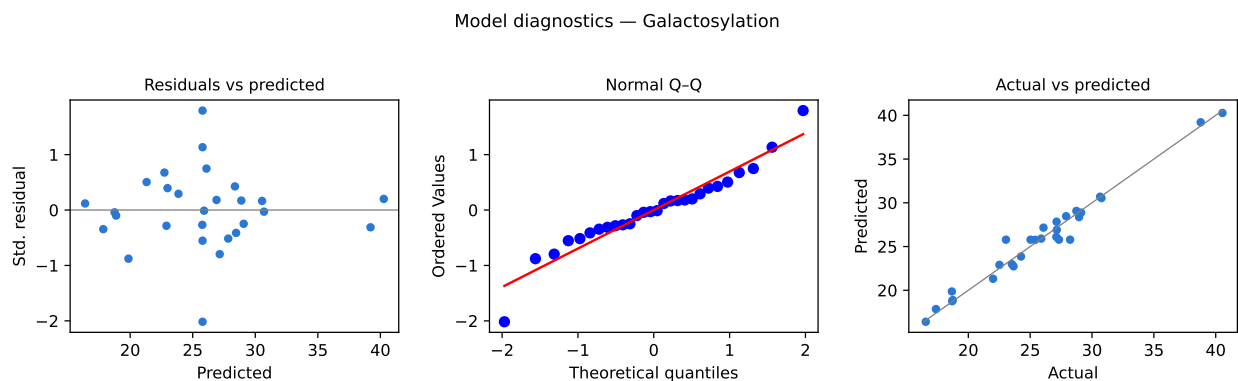


Figure 5.2: Response-surface model diagnostics for galactosylation: standardized residuals against predicted values, normal quantile plot, and actual against predicted.

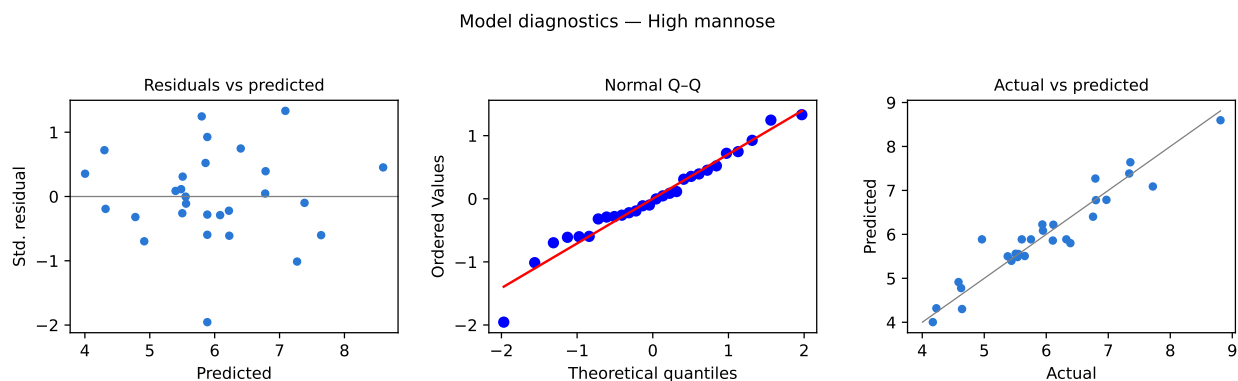


Figure 5.3: Response-surface model diagnostics for high mannose: standardized residuals against predicted values, normal quantile plot, and actual against predicted.

5.4 Mechanistic interpretation

The fitted surfaces follow the mechanisms set out in §2.1 in most respects, and they depart from them in two. Culture temperature raises galactosylation, where §2.1 expected a lower temperature to raise it, and culture pH lowers acidic charge variants, where deamidation chemistry gives the opposite sign. Both are set out below with the coefficients that carry them. The dominant effect on galactosylation is culture duration, at -3.82 per coded unit, and it is negative. Terminal galactosylation is added late in the secretory pathway by galactosyltransferase, whose activity falls as the culture ages and the intracellular environment acidifies, so material made in the last days of a long culture carries less galactose. The negative $p\text{CO}_2$ coefficient of -2.43 has the same root, since dissolved carbon dioxide lowers intracellular pH and moves the Golgi away from the optimum for that enzyme. Culture temperature acts the other way, at $+2.30$ per coded unit, so a warmer culture carries more galactose. That is the

reverse of the expectation in §2.1, which took the longer secretory residence time of a cooler culture to be the dominant route. The fitted sign says that over the range studied the temperature dependence of the transferase itself outweighs the residence time, and the screening estimate of +3.5 percentage points across the range agrees with the response-surface one.

The pH by duration interaction is the term that forces a multivariate treatment. For both afucosylation and galactosylation it is large and positive (+1.32 and +2.72 per coded unit), which means the effect of raising pH depends on how long the culture is run and can change sign across the duration range. A univariate range for pH, established at one duration, would not hold at another. This is the quantitative reason the operating region for this step is stated as a multivariate region in §6 and not as a set of independent ranges.

High mannose behaves as a simpler system. Its dominant term is culture pH at +1.09 per coded unit, with culture temperature at -0.42. High mannose species are the ones that escape processing by the mannosidases in the early Golgi, so they accumulate when processing capacity is limiting relative to secretion flux. Raising pH and lowering temperature both raise the specific secretion rate relative to that capacity, which is the expected direction and is what the model shows. Neither pCO₂ nor culture duration has a significant main effect on this response. Both interact significantly with culture pH, at +0.29 and -0.29 per coded unit (Table 5.8), so their influence is conditional on pH and is small beside the pH main effect. That pattern is consistent with a limitation set early in the pathway, in the compartment pH acts on rather than in the later ones.

Acidic charge variants are dominated by pCO₂ at -4.09 per coded unit. Deamidation of asparagine residues is base catalysed, and dissolved carbon dioxide lowers the pH of the culture fluid in which secreted antibody accumulates, so a higher pCO₂ suppresses the reaction. The pH main effect does not follow that logic. Culture pH enters the same model at -2.01 per coded unit, so a higher controlled pH gives fewer acidic variants, where base catalysis on its own would give more. The icIEF acidic peak sums species formed by more than one route, and the fitted sign is a property of that summed measurement and not of deamidation alone. The attribute is of very low criticality and the predicted levels stay well inside acceptance across the whole region (§6), so the departure is recorded here and is not carried further. Aggregate rises with pH and with duration (+0.37 and +0.28 per coded unit), which is consistent with the release of intracellular material as viability declines and with the reduced colloidal stability of the molecule at the upper end of the pH range. Neither effect is large enough to threaten the aggregate limit within the region studied, and cation exchange provides the polishing step for this attribute in any case (PCR-007).

6 Design space

The design space for the production bioreactor is a multivariate region in culture pH, culture temperature, culture duration and pCO₂, with osmolality at its normal operating range, over which every attribute the step sets remains within its acceptance criterion. It is the characterized region of Table 4.1 with one corner removed, and that single exclusion bounds everything else in this section.

One corner of the characterized region fails. With culture pH at 6.6, culture duration at 15 days and pCO₂ at 40 mmHg, all three at their low edges, and culture temperature at 36 °C at its high edge, the fitted model predicts galactosylation of 40.27 % against an upper limit of 40 %. The exceedance is 0.27 percentage points, and it is an edge of failure in the sense of the case study: a boundary the ranges were deliberately wide enough to find (CMC Biotech Working Group 2009). The mechanism is the pH by duration interaction described in §5.4, reinforced by the low pCO₂ and high temperature edges, each of which raises galactosylation on its own.

The exclusion is small and it is remote from the operating point. Across a grid of 21 levels in each of the four response-surface factors, which is 194,481 combinations, the models place all five attributes inside their acceptance criteria at all but 3 points. Every one of those points fails on galactosylation, and no other response fails anywhere in the region. Table 6.1 evaluates the same models at the 16 corners of the normal operating ranges, which is the box the process is actually controlled within.

Table 6.1: Predicted attribute levels at the 16 corners of the normal operating ranges, against the acceptance criterion for each response.

Response	Acceptance	Criterion	Min over NOR corners	Max over NOR corners
Afucosylation	2–13	two sided	6.4	9.43
Galactosylation	10–40	two sided	21.5	31.8
High mannose	3–10	two sided	5.21	6.9
Acidic variants	18–40	upper	17.7	25.7
Aggregates (HMW)	0–5	upper	1.2	1.81

Every attribute stays within its criterion at every corner of the normal operating ranges, with margin. The criterion column of Table 6.1 states how each limit applies. Acidic charge variants are judged against their upper limit alone, so the lower figure of the range stated for them is a range of experience and not an acceptance limit. The highest predicted galactosylation at any corner is 31.8 %, leaving 8.2 percentage

points to the limit, and the worst-case corner of the design space is 8.5 percentage points further out again. The three nested regions are therefore the normal operating ranges inside the design space inside the knowledge space, which is the nesting described in the case study (CMC Biotech Working Group 2009).

The plane that carries the dominant interaction is drawn in Figure 6.1, with culture temperature and pCO₂ at their set-points. No part of the excluded corner appears there, since the exclusion needs culture temperature at its high edge and pCO₂ at its low edge at the same time. The figure is the plane through the operating point and not the worst-case plane, and every point in it satisfies all five criteria.

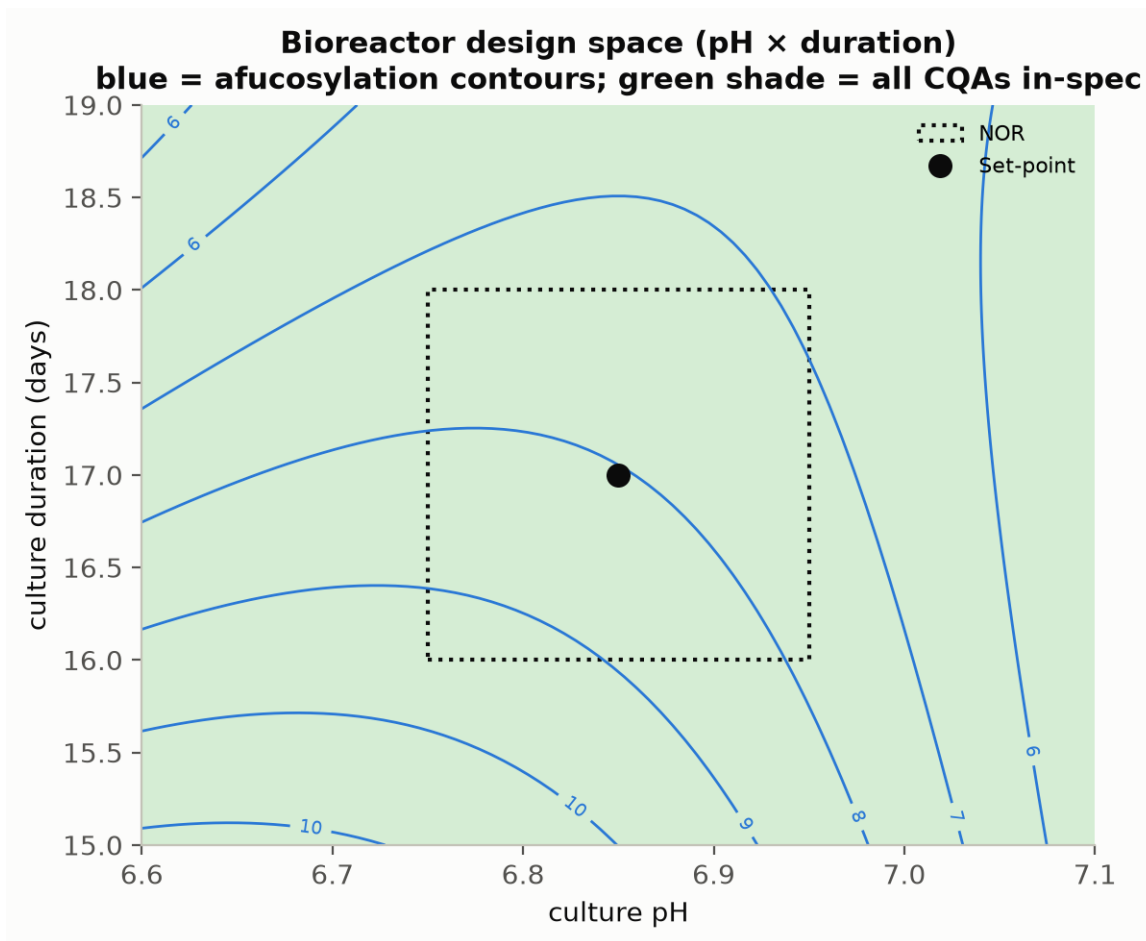


Figure 6.1: Production-bioreactor design space in the culture pH by culture duration plane, with culture temperature and pCO₂ at their set-points. The shaded region satisfies all five attribute criteria simultaneously, the contours are afucosylation, the dotted rectangle is the normal operating range and the marker is the set-point. Acceptance boundaries are drawn as dashed lines, and none falls inside this plane.

Movement within the design space is not a change from a regulatory filing perspective, as stated in ICH Q8 (International Council for Harmonisation 2009). It remains subject to the pharmaceutical quality system, so a planned move within the region is

assessed prospectively under change control before it is made (International Council for Harmonisation 2008). Movement outside the region, including any operation in the excluded corner, is a change and requires prior approval.

Three bounds apply to this claim. First, the region is defined only over the ranges characterized, and no statement is made about conditions outside Table 4.1. Second, the response-surface model predicts mean levels of each attribute, so the boundary of the region is a mean-level boundary and the assurance that a batch meets its criterion comes from the capability analysis in §8, not from the boundary itself. Third, the region rests on a scale-down model qualified for this cell line, this medium platform and these ranges, and it has not been confirmed at commercial scale at its edges. Confirmation at the set-point and within the normal operating ranges is the object of the process performance qualification.

7 Proven acceptable ranges

The proven acceptable ranges complement the design space. Where the design space is the multivariate region, a proven acceptable range is the range of one parameter over which one attribute stays within acceptance, and the two answer different questions. The acceptance criteria used are the drug-substance criteria of Table 2.1, which for the five measured attributes are the study-provided specifications and require no back-calculation. No viral clearance attribute is in scope at this step, so no step-level floor had to be derived.

Table 7.1 gives, for each governed attribute and each response-surface factor, the characterization range and both analyses described in §3.5.

Table 7.1: Proven acceptable ranges by attribute and parameter. The set-point analysis holds the other factors at their set-points; the NOR analysis varies them within their normal operating ranges by Monte-Carlo of the fitted model.

CQA	Parameter	Char. range	Unit	PAR (set-point)	PAR (NOR)
Afucosylation	Culture pH	6.6–7.1	pH	6.6–7.1	6.6–7.1
Afucosylation	Culture temperature	34–36	°C	34–36	34–36
Afucosylation	Culture duration	15–19	days	15–19	15–19
Afucosylation	Dissolved CO2 (pCO2)	40–160	mmHg	40–160	40–160
Galactosyla- tion	Culture pH	6.6–7.1	pH	6.6–7.1	6.6–7.1
Galactosyla- tion	Culture temperature	34–36	°C	34–36	34–36
Galactosyla- tion	Culture duration	15–19	days	15–19	15–19
Galactosyla- tion	Dissolved CO2 (pCO2)	40–160	mmHg	40–160	40–160
High mannose	Culture pH	6.6–7.1	pH	6.6–7.1	6.6–7.1
High mannose	Culture temperature	34–36	°C	34–36	34–36
High mannose	Culture duration	15–19	days	15–19	15–19
High mannose	Dissolved CO2 (pCO2)	40–160	mmHg	40–160	40–160

CQA	Parameter	Char. range	Unit	PAR (set-point)	PAR (NOR)
Acidic variants	Culture pH	6.6-7.1	pH	6.6-7.1	6.6-7.1
Acidic variants	Culture temperature	34-36	°C	34-36	34-36
Acidic variants	Culture duration	15-19	days	15-19	15-19
Acidic variants	Dissolved CO2 (pCO2)	40-160	mmHg	40-160	40-160
Aggregates (HMW)	Culture pH	6.6-7.1	pH	6.6-7.1	6.6-7.1
Aggregates (HMW)	Culture temperature	34-36	°C	34-36	34-36
Aggregates (HMW)	Culture duration	15-19	days	15-19	15-19
Aggregates (HMW)	Dissolved CO2 (pCO2)	40-160	mmHg	40-160	40-160

Every proven acceptable range in the table spans the full characterization range of its parameter, under both analyses. No single parameter, moved alone across the whole range studied, takes any attribute out of specification, and that remains true when the other three parameters vary randomly within their normal operating ranges. It does not mean that every combination of parameters is acceptable, and §6 shows one combination that is not. The binding constraint on the operating region at this step is the multivariate corner, not any univariate proven acceptable range.

The reason the two columns agree is that the effects are moderate relative to the width of the acceptance criteria and the normal operating ranges are narrow. Propagating normal operating variation in the other three factors widens the predictive interval of each attribute, but not far enough to reach a limit anywhere along the scanned range. The one attribute where the two analyses would be expected to separate first is galactosylation, since it has both the largest effects and the interaction that produces the excluded corner, and even there the propagated interval stays inside the criterion.

Figure 7.1 to Figure 7.5 give the analysis for each governed attribute against its governing parameter, which is the parameter with the largest fitted main effect on that attribute. The acceptable region is shaded, the acceptance limits are drawn, and the set-point and normal operating range are marked. The left panel of each figure is the set-point analysis and the right panel the Monte-Carlo analysis, with its 95 % predictive interval.

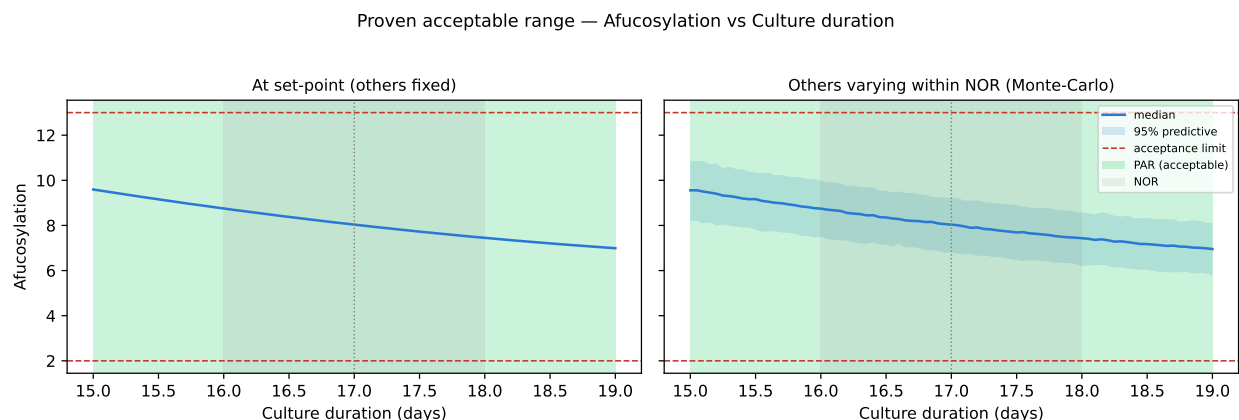


Figure 7.1: Proven acceptable range for afucosylation against its governing parameter, at the set-point of the other factors (left) and with the other factors varying within their normal operating ranges (right).

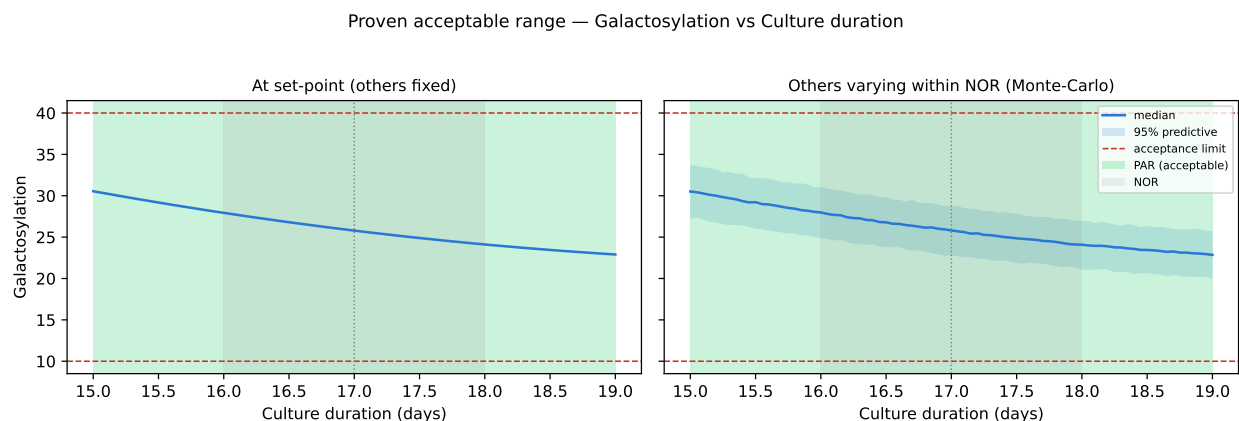


Figure 7.2: Proven acceptable range for galactosylation against its governing parameter, at the set-point of the other factors (left) and with the other factors varying within their normal operating ranges (right).

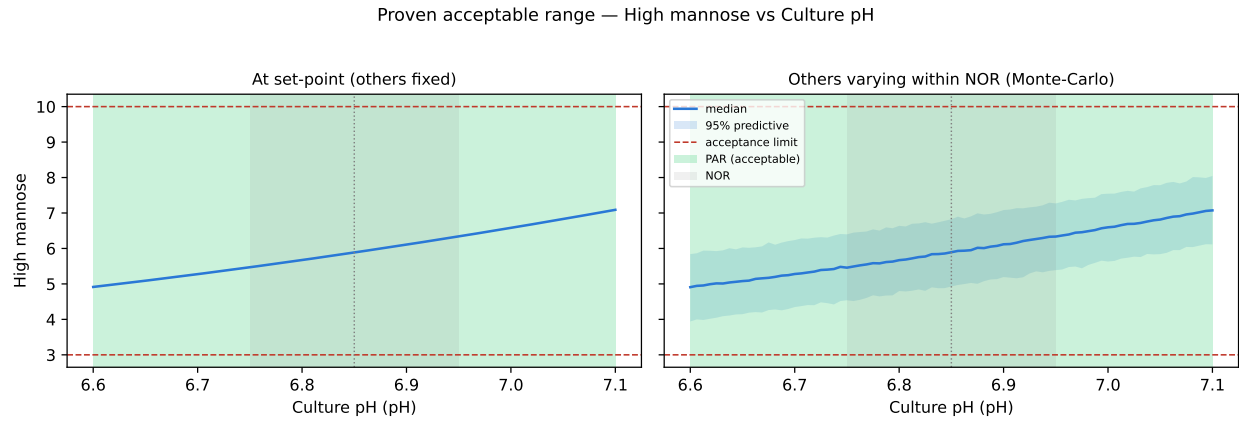


Figure 7.3: Proven acceptable range for high mannose against its governing parameter, at the set-point of the other factors (left) and with the other factors varying within their normal operating ranges (right).

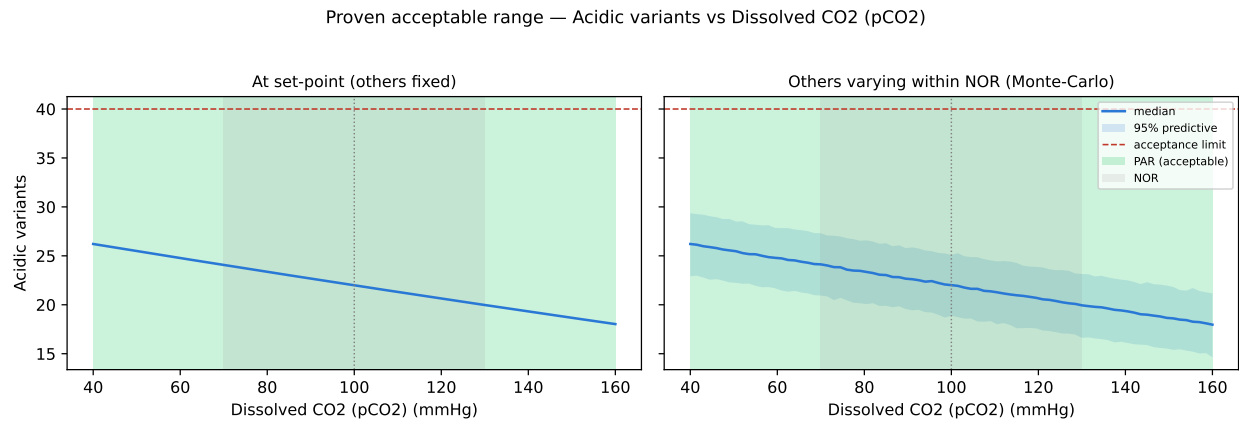


Figure 7.4: Proven acceptable range for acidic charge variants against its governing parameter, at the set-point of the other factors (left) and with the other factors varying within their normal operating ranges (right).

Proven acceptable range — Aggregates (HMW) vs Culture pH

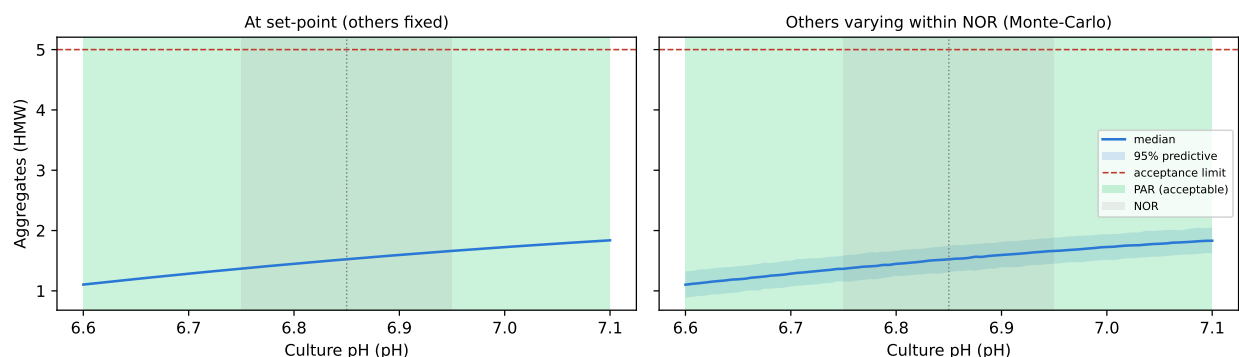


Figure 7.5: Proven acceptable range for high molecular weight aggregate against its governing parameter, at the set-point of the other factors (left) and with the other factors varying within their normal operating ranges (right).

Two bounds apply. The ranges are proven over the characterization ranges of Table 4.1 and against the fitted response-surface model, so they are supported by data only where data were taken. They are also computed at osmolality within its normal operating range, since osmolality was not a response-surface factor (§4.3).

8 Process capability and robustness

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches, each with every parameter drawn independently within its normal operating range and each attribute predicted from the fitted models with their residual variability. Table 8.1 gives the outcome for the seven attributes the step sets.

Table 8.1: Predicted commercial-scale capability for the attributes set by the production bioreactor, from 2,000 simulated batches with all parameters inside their normal operating ranges.

CQA	Crit.	Spec	Acceptance	Mean $\pm$ SD	Cpk
Afucosylation	H	two_sided	2-13	7.98 ± 0.58	2.89
Galactosylation (total %Gal)	H	two_sided	10-40	26 ± 1.6	2.86
High Mannose	VH	two_sided	3-10	5.99 ± 0.51	1.94
Aggregates (HMW)	H	upper	0-5	0.753 ± 0.088	16.1
Acidic charge variants (deamidation)	VL	upper	18-40	22.3 ± 1.4	4.26
Host Cell Protein (HCP)	M-H	upper	0-100	15.2 ± 4.6	6.14
Residual DNA	VL	upper	0-0.001	0.0001 ± 0	10.8

Every attribute meets its acceptance criterion, and the tightest capability is comfortable. High Mannose is the binding attribute with a one-sided Cpk of 1.94. Its predicted mean is 5.99 % of glycans, which sits 2.99 percentage points above the lower limit and 4.01 percentage points below the other one. That this is the tightest of the seven follows from the attribute and not from the process, since high mannose has the narrowest acceptance range of the glycan attributes and the largest relative variability at the centre point (\$5.1). The other two glycan attributes follow at Cpk values close to 2.9.

Three of the seven attributes carry capability that is not this step's alone. Aggregates (HMW) shows the highest index in the table at 16, which reflects the cation exchange polish and not the culture (PCR-007). The bioreactor delivers aggregate at the level shown at the centre point in Table 5.1, and the drug-substance mean is lower than that by a factor of 2.0. Host cell protein and residual DNA are generated in the culture at levels orders of magnitude above their drug substance criteria, and the indices shown here are the product of clearance at Protein A, cation exchange and anion exchange (PCR-005, PCR-007, PCR-008). For those three attributes the contribution of this step

is the input level to the purification train, and that input level is what the culture parameters control.

The capability estimate is bounded in two ways. It assumes every parameter stays inside its normal operating range, which is what the control system delivers and what the parameter classifications in §9 commit to. It also rests on the qualified scale-down model, since the underlying models were fitted to scale-down data. Confirmation at commercial scale is the object of the process performance qualification and is not claimed here.

9 Parameter classification

Each parameter was classified per SOP-4001 from the effects demonstrated in this study and from the risk of the parameter leaving the design space in routine operation. A parameter whose variability affects a critical quality attribute is a critical process parameter, and it is called well controlled when the equipment and the control scheme make an excursion outside the design space unlikely (CMC Biotech Working Group 2009). The classifications are given in Table 9.1 and the reasoning follows.

Table 9.1: Final classification of every production-bioreactor parameter, with the ranges that support it.

Parameter	Unit	Set-point	NOR	Char. range	Study	Class
Culture pH	pH	6.85	6.75–6.95	6.6–7.1	multivari- ate	WC- CPP
Culture temperature	°C	35	34.5–35.5	34–36	multivari- ate	WC- CPP
Dissolved CO ₂ (pCO ₂)	mmHg	100	70–130	40–160	multivari- ate	WC- CPP
Osmolality	mOsm	400	375–425	360–440	multivari- ate	WC- CPP
Culture duration	days	17	16–18	15–19	multivari- ate	WC- CPP
Dissolved oxygen	% sat	50	40–60	30–70	univari- ate	KPP
Initial viable cell conc.	e6 cells/mL	1	0.8–1.2	0.5–1.5	univari- ate	KPP
Basal medium concentration	X	1.2	1–1.4	0.8–1.6	univari- ate	GPP
Nutrient feed-1 volume	% WV	12	10–14	9–15	univari- ate	KPP

- Culture pH is a WC-CPP. It has a significant effect on four of the five measured attributes and it is the governing parameter for high mannose and for aggregate. It is held by closed-loop control with redundant calibrated probes, and the normal operating range is a fraction of the characterized range, so the risk of leaving the design space is low.
- Culture temperature is a WC-CPP. It affects galactosylation, high mannose and acidic variants, and it interacts with pH. Jacket control holds the culture well within the normal operating range and an excursion would be detected within minutes.

- pCO₂ is a WC-CPP. It carries the largest single effect in the screening study, on acidic charge variants, and a substantial effect on galactosylation. It is the least tightly controlled of the five multivariate parameters, since it depends on sparge efficiency and on vessel scale, which is why the characterized range is wide. The classification is well controlled and not critical because the proven acceptable range for this parameter spans its full characterized range under both analyses (§7), and an excursion beyond that range is not physically plausible in the commercial vessel.
- Osmolality is a WC-CPP. It has significant effects on galactosylation and on acidic variants in the screening study, and it was not carried into the response-surface design (§4.3). It is classified from the screening result and is controlled by the feed and base strategy, with at-line monitoring through the culture.
- Culture duration is a WC-CPP. It carries the largest effect on galactosylation and the largest main effect on afucosylation, and it is one of the two parameters in the dominant interaction. It is set by the harvest decision and not by a controller, and the harvest criteria in SOP-2003 bound it directly, so the risk of operating outside the range is low.
- Dissolved oxygen is a KPP. It drives peak viable cell density and titre and no effect of it on a quality attribute has been demonstrated. It is monitored and controlled because process consistency depends on it, which is the purpose a key process parameter serves.
- Initial viable cell concentration is a KPP, on the same basis. It sets the growth trajectory and the integral of viable cells, and therefore titre, without a demonstrated quality effect.
- Basal medium concentration is a GPP. It affects titre, no effect of it on a quality attribute was demonstrated over the range studied (§4.4), and it is controlled by the media preparation procedure.
- Nutrient feed-1 volume is a KPP. It is critical to titre and is delivered gravimetrically under the control described in §10, following the deviation reported in §13.2.

The step has 5 well-controlled critical process parameters, 3 key process parameters and 1 general process parameter. No parameter of this step is classified as a critical process parameter. That outcome rests on control capability rather than on the absence of effects, since four of the five multivariate parameters have large demonstrated effects. The commercial vessel holds each of them within a normal operating range that sits well inside the design space, and the capability analysis in §8 quantifies what that containment delivers.

10 Contribution to the control strategy

This step contributes the elements of the control strategy that govern the formation of the glycan, charge variant and aggregate attributes. They are as follows.

- The five multivariate parameters are controlled to the normal operating ranges in Table 4.1, and the design space of §6 defines the region within which they may be moved under the quality system.
- Culture pH, temperature, dissolved oxygen and pCO₂ are monitored continuously by calibrated in-process instrumentation, and osmolality is monitored at line. Viable cell density, viability and titre are trended daily against the platform profile given in §3.2.
- Harvest is bounded by the criteria in SOP-2003, which controls culture duration directly and is the control that keeps the aggregate and host cell protein input to the purification train within the characterized range.
- Nutrient feed-1 is delivered gravimetrically with mass reconciliation after each addition, and the feed pump is verified before each campaign. This control was strengthened after DEV-003-02 (§13.2).
- Released N-glycan mapping, size variants, charge variants, host cell protein and residual DNA are tested on the drug substance under the methods of Table 3.1. Testing confirms the outcome; it does not control it, and the control rests on the parameter ranges above.
- The scale-down model is maintained under SOP-1001 so that any future range change can be assessed on a qualified system and not at scale.

The step does not deliver the drug substance specification for aggregate, host cell protein or residual DNA on its own. What it delivers for those three attributes is a bounded input to the purification train, and the clearance that turns that input into the drug substance levels of Table 8.1 is characterized in PCR-005, PCR-007 and PCR-008. The overall control strategy, including how the elements of this step combine with the downstream ones, is consolidated in PCMR-001.

11 Discussion

The step is well understood over the ranges characterized, and the understanding is mechanistic and not merely empirical. Most of the significant main effects in Table 5.3 run through a route in the secretory pathway or in culture chemistry that was set out before the study was run (§2.1), and the fitted models reproduce those routes in direction. Two main effects do not. Culture temperature raises galactosylation where a lower temperature was expected to raise it, and culture pH lowers acidic charge variants where deamidation chemistry gives the opposite sign, and §5.4 sets both out. The study also returned significant interactions that §2.1 did not anticipate, since prior knowledge carried the pH by duration term alone. What the study added there was the size of that term, which is what makes a multivariate operating region necessary rather than convenient.

The adverse finding of the study is the excluded corner. At the low edges of pH, duration and pCO₂ with temperature high, galactosylation is predicted at 40.27 % against a limit of 40 %. That corner is excluded from the design space, and no operation there is permitted without prior approval. Two things bound its practical significance. It requires four parameters to sit simultaneously at the edges of ranges that are between 2.0 and 2.5 times the width of their normal operating ranges, and the normal operating ranges themselves leave 8.2 percentage points of margin at their worst corner (Table 6.1). The finding does not weaken the design space claim. It is the reason the claim can be stated as a region rather than as an assumption.

Confidence in the scale-down model is good but not unlimited. The model is geometrically similar to the commercial vessel and matches it on the scale-independent parameters, and the scale-dependent parameters were set on performance and not on setting (§3.1). Its qualification covers this cell line, this medium platform and these ranges. Two known differences remain. Carbon dioxide accumulates less readily at small scale than in a commercial vessel, which is the reason the pCO₂ range was carried to 160 mmHg instead of stopping at the range seen at laboratory scale. Mixing time also scales with vessel size, so gradients that are absent at small scale can exist at commercial scale. Both are managed by the process performance qualification, which measures the same attributes on commercial batches.

Three further limitations should be read with the conclusions. The response-surface model predicts mean levels and its edges carry no assurance statement on their own, which is why the capability analysis exists (§5.3, §8). Osmolality was characterized in screening only, so its contribution to the region is bounded by a two-level estimate and is not described by a surface (§4.3). The univariate parameters were assessed against process performance, so the quality claims of this report rest on the multivariate work alone (§4.4). None of the three affects the classification of any parameter.

Two deviations occurred during execution and both were retained. A pCO₂ probe drifted between calibrations on one screening run, and one centre-point batch received less nutrient feed than the schedule specifies. The magnitudes are small against the ranges studied, the impact assessments are in §13, and neither changed an effect estimate, a classification or the boundary of the operating region. No parameter of this step required the most stringent level of control, and the argument for that is control capability rather than an absence of effects (§9).

12 Conclusions

The production bioreactor is characterized and the step is robust over the ranges studied. A response-surface model over culture pH, culture temperature, culture duration and pCO₂ defines a multivariate design space in which the five measured attributes remain within their acceptance criteria, with one corner of the characterized region excluded on galactosylation. The normal operating ranges lie well inside that region, with 8.2 percentage points of margin on the binding attribute at the worst corner of the operating box.

Proven acceptable ranges span the full characterization range of every response-surface parameter against every attribute it governs, under both the set-point and the normal-operating-range analyses, so the constraint that binds the operating region is the multivariate corner and not a univariate range. Predicted commercial-scale capability meets acceptance for all seven attributes the step sets, with High Mannose the tightest at a one-sided Cpk of 1.94 and 2.99 percentage points to its binding limit. The study classified 9 parameters, 5 of them as well-controlled critical process parameters and none as a critical process parameter. Two deviations were investigated and resolved without impact on any of these outcomes.

Two claims are not made here. The design space has not been confirmed at commercial scale at its edges, and confirmation within the normal operating ranges is the object of the process performance qualification. The drug substance levels of aggregate, host cell protein and residual DNA depend on downstream clearance as well as on culture conditions, and the credit for that clearance belongs to PCR-005, PCR-007 and PCR-008. The outcomes of this report are consolidated with those of the other steps in PCMR-001.

13 Deviations from the plan

Two deviations from PCP-003 were recorded during execution. Each was raised at the time of detection, investigated to root cause, assessed for impact on the study conclusions and closed. Both were dispositioned as retained, meaning the affected run stayed in the analysis. Table 13.1 is the register.

Table 13.1: Deviations recorded during execution of the production-bioreactor characterization.

Devia- tion	Summary	Detected during	Disposi- tion
DEV-003-01	Dissolved-CO ₂ probe drift during screening run 4	in-process monitoring vs offline reference	retained
DEV-003-02	Nutrient feed-1 under-delivery on a centre-point batch	post-execution mass reconciliation	retained

The two entries are of different kinds. The first is an instrument drift affecting one factor setting on one screening run, and the second is a materials delivery shortfall on a centre-point batch. Neither affected a response-surface run. The subsections below give the adverse finding, the investigation and the impact assessment for each.

13.1 DEV-003-01 — dissolved carbon dioxide probe drift

The in-process pCO₂ probe read 6.5 mmHg below the offline reference on screening run 4, so that run was executed at a true pCO₂ above its assigned setting. The discrepancy was found by the routine comparison of the in-process reading against an offline blood gas measurement, and it was confirmed on the retained sample.

The investigation attributed the discrepancy to drift of the probe between scheduled calibrations. The probe is part of instrumented system EQ-BRX-205, whose calibration status at the time of execution is given in Table 13.2. The probe was within its calibration interval when the run was executed, which is why the drift was not detected by the calibration programme, and it was recalibrated against the offline reference before the response-surface design was executed. No other run in either design showed a comparable discrepancy.

Table 13.2: Calibration record of the instrumented scale-down system affected by DEV-003-01.

Equip- ment	Description	Calibration status	Next calibration due
EQ-BRX- 205	production scale-down bioreactor pCO ₂ probe	in calibration at execution	2026-06-14

The impact assessment bounds the consequence on the fitted effects. The offset corresponds to 0.11 coded units on the pCO₂ axis, which is a small fraction of the distance from the set-point to the edge of the characterization range. Evaluating the fitted response-surface model at the set-point with pCO₂ displaced by that amount moves the most pCO₂-sensitive response, acidic variants, by 0.44 percentage points. That shift is smaller than the centre-point standard deviation of the same response, 0.88 percentage points (Table 5.1), so it is not resolvable against the reproducibility of the study. The run was retained. The true setting remained inside the characterization range of pCO₂, the direction of every screening effect is unchanged, and the design space of §6 rests on the response-surface design, which the deviation does not touch.

13.2 DEV-003-02 — nutrient feed-1 under-delivery

One centre-point batch of the screening design received 4.2 % less nutrient feed-1 than the schedule specifies, which is a delivered volume of 11.5 % of working volume against a set-point of 12 %. The shortfall was found after execution by mass reconciliation of feed lot LOT-FED-3120 against the batch record, not by an in-process alarm.

Root cause was a calibration offset on the feed pump, which delivered consistently below its commanded volume across the campaign period. The pump was recalibrated and the delivery verification described in §10 was introduced as a preventive action. The shortfall is a materials deviation and not a parameter excursion in the design, since nutrient feed-1 volume is a univariate parameter and was held at its set-point in every designed run.

The impact assessment compared the affected batch against 3 verification runs executed at the centre point after the pump was recalibrated, with the released glycan attributes measured by AMV-3010. The half width of the 95 % confidence interval on the difference is 6.21 % relative, and the observed difference lies inside it, so no difference in the glycan profile was demonstrated. The delivered volume of 11.5 % of working volume also remains inside the normal operating range of that parameter, which is 10 to 14 % (Table 4.1).

The batch was retained as a centre point. Retaining it slightly inflates the pure-error term of the screening design, which is conservative, since a larger pure error makes a lack-of-fit test easier to pass and the screening design is not used for prediction in any case (§5.2). The lack-of-fit tests that support the design space come from the

response-surface design, whose 4 centre points were unaffected. No effect estimate, no classification and no boundary of the operating region was changed by this deviation.

Appendix A — Screening design matrix and responses

The coded settings of the 19 screening runs are given in Table 13.3 and the measured responses in Table 13.4. Factor codes are given in Table 4.2, and the coded levels minus one, zero and plus one correspond to the low edge, the set-point and the high edge of each characterization range.

Table 13.3: Screening design, coded factor settings by run.

Run	Type	A	B	C	D	E
1	factorial	-1	-1	-1	-1	1
2	factorial	-1	-1	-1	1	-1
3	factorial	-1	-1	1	-1	-1
4	factorial	-1	-1	1	1	1
5	factorial	-1	1	-1	-1	-1
6	factorial	-1	1	-1	1	1
7	factorial	-1	1	1	-1	1
8	factorial	-1	1	1	1	-1
9	factorial	1	-1	-1	-1	-1
10	factorial	1	-1	-1	1	1
11	factorial	1	-1	1	-1	1
12	factorial	1	-1	1	1	-1
13	factorial	1	1	-1	-1	1
14	factorial	1	1	-1	1	-1
15	factorial	1	1	1	-1	-1
16	factorial	1	1	1	1	1
17	center	0	0	0	0	0
18	center	0	0	0	0	0
19	center	0	0	0	0	0

Table 13.4: Screening design, measured responses by run.

Run	Type	Afucosylation	Galactosylation	High mannose	Acidic variants	Aggregates (HMW)
1	factorial	5.18	27.9	6.61	26.4	1.3
2	factorial	8.93	36.1	6.44	26.5	0.783

Run	Type	Afucosylation	Galactosylation	High mannose	Acidic variants	Aggregates (HMW)
3	factorial	11.8	34.6	5.63	22.3	0.785
4	factorial	4.83	14.7	5.46	17.9	1.48
5	factorial	9.72	41.4	4.61	27	0.859
6	factorial	4.68	27.3	4.68	27.6	1.26
7	factorial	6.22	14.5	4.65	23.1	1.48
8	factorial	9.83	31.2	4.45	19.3	0.948
9	factorial	9.61	28	7.77	24.1	1.51
10	factorial	7.51	17	7.89	20.4	2.26
11	factorial	7.13	19.5	7.34	12.6	2.53
12	factorial	5.29	17.6	7.01	11.3	1.61
13	factorial	6.74	28.3	6.43	29.6	2.18
14	factorial	5.4	26.7	5.94	24.4	1.62
15	factorial	3.82	25.2	6.1	18.4	1.8
16	factorial	3.08	29.1	6.15	15.8	2.05
17	center	7.8	26.4	5.88	21.9	1.64
18	center	7.51	27.4	5.84	20	1.63
19	center	8.48	27	7.11	21.8	1.5

Appendix B — Response-surface design matrix and responses

The coded settings of the 28 response-surface runs are given in Table 13.5 and the measured responses in Table 13.6. Runs of type factorial are the cube points, axial runs sit on the faces of the cube, and centre runs are at the set-point of every factor.

Table 13.5: Response-surface design, coded factor settings by run.

Run	Type	A	B	E	C
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

Table 13.6: Response-surface design, measured responses by run.

Run	Type	Afucosylation	Galactosylation	High mannose	Acidic variants	Aggregates (HMW)
1	factorial	9.43	38.8	5.44	29.8	0.827
2	factorial	11.8	29	5.55	17.6	0.791
3	factorial	4.5	25.9	5.65	25.1	1.33
4	factorial	5.38	16.6	5.38	20	1.37
5	factorial	10.4	40.5	4.23	25.6	0.803
6	factorial	10.1	30.7	4.17	19.8	0.707
7	factorial	5.12	27.1	4.63	30.3	1.24
8	factorial	6.13	17.4	4.64	21.8	1.41
9	factorial	8.92	22	6.79	22.9	1.5
10	factorial	5.89	18.7	8.81	13.6	1.53
11	factorial	8.05	18.7	6.11	19.3	1.99
12	factorial	5.17	18.7	7.34	12.3	2.15
13	factorial	5.41	28.7	6.97	28.5	1.45
14	factorial	2.84	29.1	7.36	18.5	1.7
15	factorial	6.2	25.4	5.95	26.5	2.08
16	factorial	4.19	27.2	6.8	17.2	2.01
17	axial	7.82	26.1	4.59	24.2	1.15
18	axial	6.07	23.7	7.72	19.3	1.82
19	axial	8.77	24.3	6.75	19.9	1.51
20	axial	7.55	27.9	5.51	26.8	1.32
21	axial	10.3	30.8	6.11	20.8	1.3
22	axial	6.89	22.5	5.54	22.4	2.06
23	axial	8.48	27.1	6.39	25.7	1.5
24	axial	8.39	23.5	5.94	19.2	1.45
25	center	7.88	23	5.75	20.5	1.52
26	center	7.95	27.3	6.32	22	1.6
27	center	7.35	25	5.6	22.4	1.55

Run	Type	Afucosylation	Galactosylation	High mannose	Acidic variants	Aggregates (HMW)
28	center	7.27	28.2	4.96	21.1	1.37

Appendix C — Full effect and coefficient tables

The complete screening effect tables for all 5 responses are given first, followed by the response-surface coefficient tables for the three responses not shown in §5.3. Terms are ordered by the absolute size of the effect in the screening tables and in model order in the coefficient tables.

Afucosylation — screening effects

Table 13.7: Screening effect estimates, Afucosylation.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AE	2.46	1.23	0.266	4.62	0.0191	*
E	-2.37	-1.19	0.266	-4.46	0.0209	*
AC	-1.76	-0.88	0.266	-3.31	0.0455	*
A	-1.57	-0.785	0.266	-2.95	0.06	
B	-1.34	-0.671	0.266	-2.53	0.0858	
D	-1.33	-0.664	0.266	-2.5	0.0878	
AB	-1.28	-0.64	0.266	-2.41	0.0952	
C	-0.73	-0.365	0.266	-1.37	0.264	
BD	0.452	0.226	0.266	0.85	0.458	
BE	0.362	0.181	0.266	0.681	0.545	
AD	-0.175	-0.0876	0.266	-0.329	0.764	
BC	-0.171	-0.0856	0.266	-0.322	0.769	
CD	-0.147	-0.0736	0.266	-0.277	0.8	
DE	0.0358	0.0179	0.266	0.0674	0.951	
CE	0.0118	0.00591	0.266	0.0222	0.984	

Galactosylation — screening effects

Table 13.8: Screening effect estimates, Galactosylation.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
E	-7.82	-3.91	0.193	-20.2	0.000264	***
AE	6.9	3.45	0.193	17.9	0.000382	***
C	-5.79	-2.9	0.193	-15	0.000644	***
A	-4.56	-2.28	0.193	-11.8	0.00131	**
BD	3.7	1.85	0.193	9.57	0.00242	**
AC	3.6	1.8	0.193	9.32	0.00261	**

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	3.54	1.77	0.193	9.17	0.00274	**
AB	3.28	1.64	0.193	8.49	0.00343	**
D	-2.45	-1.23	0.193	-6.35	0.0079	**
CD	2.18	1.09	0.193	5.64	0.011	*
DE	1.93	0.963	0.193	4.98	0.0155	*
BE	1.49	0.747	0.193	3.87	0.0306	*
AD	-0.195	-0.0974	0.193	-0.504	0.649	
BC	-0.131	-0.0657	0.193	-0.34	0.756	
CE	0.119	0.0596	0.193	0.309	0.778	

High mannose — screening effects

Table 13.9: Screening effect estimates, High mannose.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1.51	0.756	0.155	4.89	0.0164	*
B	-1.39	-0.696	0.155	-4.5	0.0205	*
C	-0.446	-0.223	0.155	-1.44	0.245	
BC	0.369	0.185	0.155	1.19	0.318	
E	0.158	0.0791	0.155	0.511	0.645	
D	-0.139	-0.0693	0.155	-0.448	0.685	
AC	0.0897	0.0449	0.155	0.29	0.791	
AE	0.0894	0.0447	0.155	0.289	0.792	
DE	-0.0718	-0.0359	0.155	-0.232	0.831	
CE	-0.0552	-0.0276	0.155	-0.179	0.87	
BE	0.046	0.023	0.155	0.149	0.891	
AB	0.0432	0.0216	0.155	0.14	0.898	
AD	-0.0239	-0.012	0.155	-0.0774	0.943	
CD	-0.0232	-0.0116	0.155	-0.075	0.945	
BD	-0.000576	-0.000288	0.155	-0.00186	0.999	

Acidic variants — screening effects

Table 13.10: Screening effect estimates, Acidic variants.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
C	-8.15	-4.08	0.239	-17	0.000441	***
A	-4.17	-2.08	0.239	-8.7	0.00319	**
B	2.97	1.48	0.239	6.19	0.00848	**
D	-2.55	-1.28	0.239	-5.34	0.0129	*
AB	1.98	0.992	0.239	4.15	0.0255	*
AC	-1.93	-0.964	0.239	-4.03	0.0275	*
BE	1.73	0.865	0.239	3.61	0.0364	*

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AD	-0.669	-0.335	0.239	-1.4	0.256	
CE	-0.494	-0.247	0.239	-1.03	0.378	
CD	-0.492	-0.246	0.239	-1.03	0.379	
BD	-0.215	-0.108	0.239	-0.449	0.684	
BC	0.151	0.0757	0.239	0.316	0.773	
DE	0.0324	0.0162	0.239	0.0676	0.95	
E	0.0235	0.0117	0.239	0.049	0.964	
AE	0.0226	0.0113	0.239	0.0472	0.965	

Aggregates (HMW) — screening effects

Table 13.11: Screening effect estimates, Aggregates (HMW).

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	0.834	0.417	0.0207	20.1	0.000268	***
E	0.578	0.289	0.0207	13.9	0.0008	***
BE	-0.141	-0.0705	0.0207	-3.4	0.0424	*
C	0.116	0.058	0.0207	2.8	0.0678	
CD	-0.0742	-0.0371	0.0207	-1.79	0.171	
AD	-0.0666	-0.0333	0.0207	-1.61	0.206	
AB	-0.0593	-0.0297	0.0207	-1.43	0.248	
DE	-0.0554	-0.0277	0.0207	-1.34	0.274	
D	-0.0548	-0.0274	0.0207	-1.32	0.278	
BD	-0.0547	-0.0274	0.0207	-1.32	0.278	
AE	0.0416	0.0208	0.0207	1	0.39	
BC	-0.0235	-0.0118	0.0207	-0.567	0.61	
CE	0.0222	0.0111	0.0207	0.537	0.629	
B	-0.00671	-0.00336	0.0207	-0.162	0.882	
AC	-0.00663	-0.00332	0.0207	-0.16	0.883	

Afucosylation — response-surface coefficients

Table 13.12: Response-surface coefficients on coded factors, Afucosylation.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	8.04	0.202	39.8	5.74e-15	***
A	-0.997	0.138	-7.24	6.52e-06	***
B	-0.551	0.138	-4	0.00151	**
E	-1.3	0.138	-9.45	3.45e-07	***
C	-0.367	0.138	-2.67	0.0194	*
AB	-0.63	0.146	-4.32	0.000837	***
AE	1.32	0.146	9.06	5.6e-07	***
AC	-0.907	0.146	-6.21	3.16e-05	***

Term	Coef.	Std. err.	t	p-value	Sig.
BE	0.362	0.146	2.48	0.0277	*
BC	-0.0703	0.146	-0.481	0.638	
EC	0.0322	0.146	0.221	0.829	
A ²	-1.37	0.364	-3.77	0.00233	**
B ²	-0.157	0.364	-0.431	0.673	
E ²	0.257	0.364	0.705	0.493	
C ²	0.117	0.364	0.323	0.752	

Acidic variants — response-surface coefficients

Table 13.13: Response-surface coefficients on coded factors, Acidic variants.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	22	0.542	40.6	4.43e-15	***
A	-2.01	0.369	-5.44	0.000113	***
B	1.9	0.369	5.14	0.000191	***
E	-0.124	0.369	-0.335	0.743	
C	-4.09	0.369	-11.1	5.4e-08	***
AB	1.11	0.392	2.84	0.0139	*
AE	-0.793	0.392	-2.03	0.0638	
AC	-0.248	0.392	-0.633	0.538	
BE	0.655	0.392	1.67	0.118	
BC	-0.0111	0.392	-0.0283	0.978	
EC	0.465	0.392	1.19	0.256	
A ²	-0.586	0.975	-0.601	0.558	
B ²	1.03	0.975	1.06	0.311	
E ²	-0.718	0.975	-0.736	0.475	
C ²	0.13	0.975	0.133	0.896	

Aggregates (HMW) — response-surface coefficients

Table 13.14: Response-surface coefficients on coded factors, Aggregates (HMW).

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	1.52	0.0341	44.8	1.25e-15	***
A	0.367	0.0232	15.8	7.26e-10	***
B	-0.0157	0.0232	-0.675	0.512	
E	0.281	0.0232	12.1	1.91e-08	***
C	0.0225	0.0232	0.971	0.349	
AB	0.0135	0.0246	0.547	0.593	
AE	-0.0106	0.0246	-0.43	0.674	
AC	0.0179	0.0246	0.727	0.48	
BE	-0.00713	0.0246	-0.29	0.777	

Term	Coef.	Std. err.	t	p-value	Sig.
BC	0.00414	0.0246	0.168	0.869	
EC	0.0105	0.0246	0.426	0.677	
A ²	-0.0533	0.0613	-0.87	0.4	
B ²	-0.119	0.0613	-1.93	0.0752	
E ²	0.143	0.0613	2.33	0.0366	*
C ²	-0.0626	0.0613	-1.02	0.326	

Appendix D — Analytical methods summary

Table 13.15 summarizes the validated methods used to measure the attributes reported here. Performance figures are given where a campaign method-performance record exists, and the remainder are stated in the method validation reports listed in Table 3.1.

Table 13.15: Analytical methods, the attributes they report and their validated performance.

Ref- er- ence	Method	Attribute reported	Re- portable unit	Precision (%RSD)	LOQ
AMV- 3010	N-Glycan Map (2-AB HILIC-UPLC)	Afucosylation, Galactosylation (total %Gal), High Mannose	% of glycans	2.5	0.5 % glycan
AMV- 3011	Size-Variants (SEC-HPLC)	Aggregates (HMW)	% HMW (SEC)	per validation report	per validation report
AMV- 3012	Host-Cell Protein ELISA	Host Cell Protein (HCP)	ng/mg	9.5	1 ng/mg
AMV- 3013	Charge Variants (icIEF)	Acidic charge variants (deamidation)	% (CEX/icIEF)	per validation report	per validation report
AMV- 3014	Residual DNA (qPCR)	Residual DNA	ng/dose	per validation report	per validation report

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